Aerodynamic Design Analysis and Fabrication of a UAV Test Model Capable of Delivering Medical Supplies

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AIAA SHAPING THE FUTURE OF AEROSPACE



CONTEXT OF RESEARCH AND GOALS

Current Global Challenges



1

COVID-19 PANDEMIC

An unprecedented viral outbreak that ravaged the global health system and revealed its shortcomings.

2



CLIMATE CHANGE

Global warming is, by far, the biggest concern, as we are running out of time to save the EARTH.

3



ECONOMIC RECESSION

The shutdown of worldwide major economies during the pandemic has yet again brought us on the verge of another great recession after 2008.

Our Goals

GOAL To design and fabricate a cargo unmanned aircraft, addressing three global aspects.

1.

MEDICAL AID



Capability to deliver medical goods, such as vaccination components to inaccessible areas.

2.

GREEN ENERGY



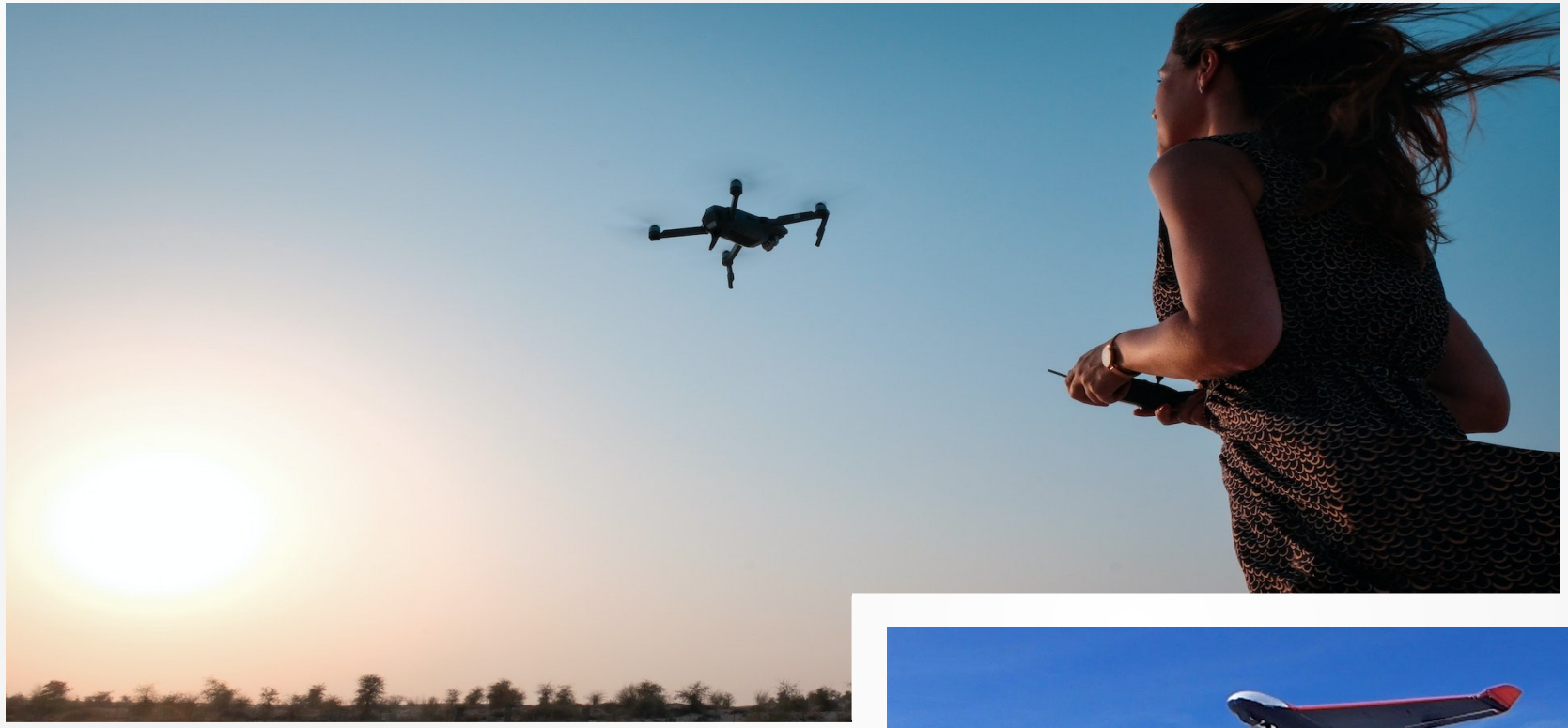
Use of sustainable energy as the major source of propulsion.

3.

COST EFFICIENT



Application of materials and fabrication techniques that are cost-effective, as well as reliable in quality.

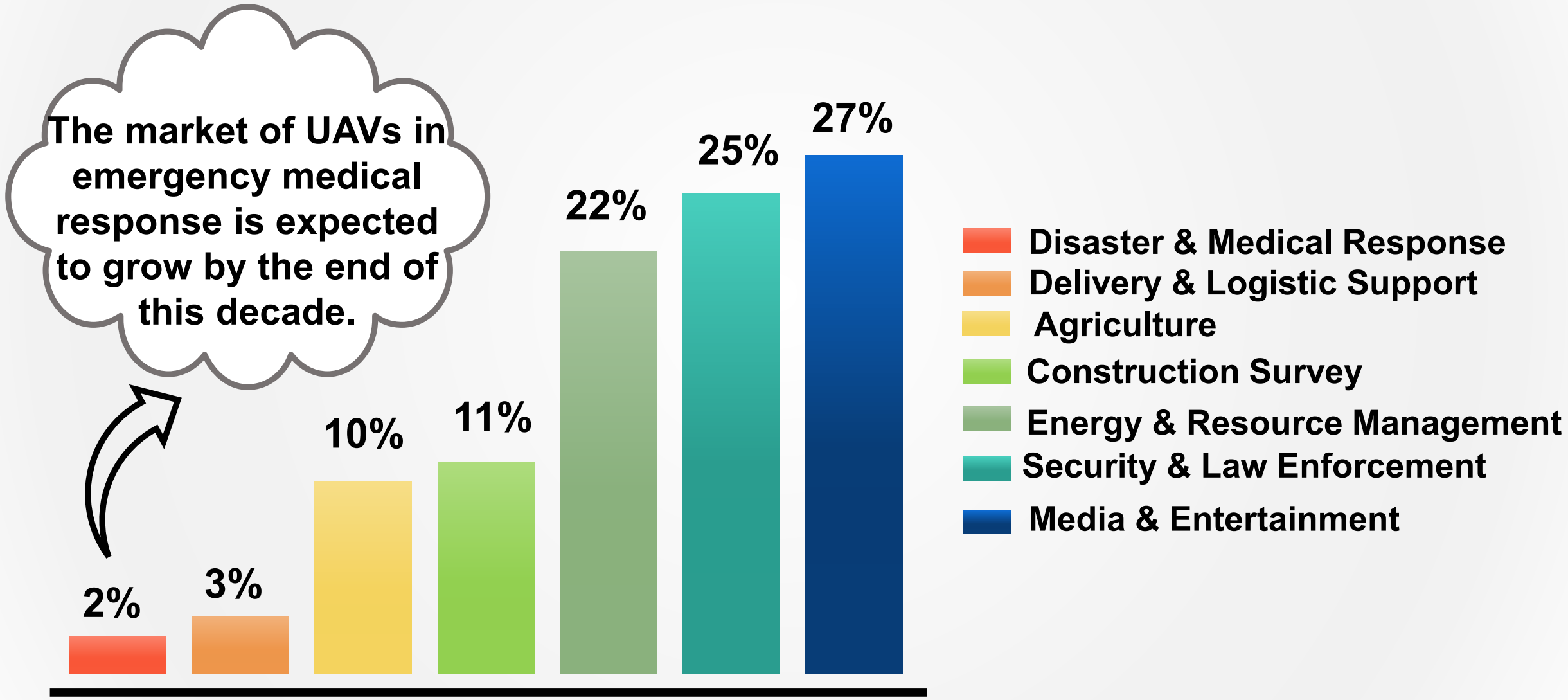




CURRENT TRENDS AND FUTURE OF UAV



UAV Market **Share by Areas of Applications in (%)**



Fixed-Wing Vs. Rotary Drone



PROS

- ❑ Capability to fly at higher altitudes.
- ❑ Can travel great distances at high speeds.
- ❑ Significantly larger payload capacity.

CONS

- ❑ Requires runway

PROS

- ❑ Has the capability to take off and land vertically.

CONS

- ❑ Shorter flying range with very lower speed.
- ❑ Smaller payload capacity



Project SE-UAV Overview

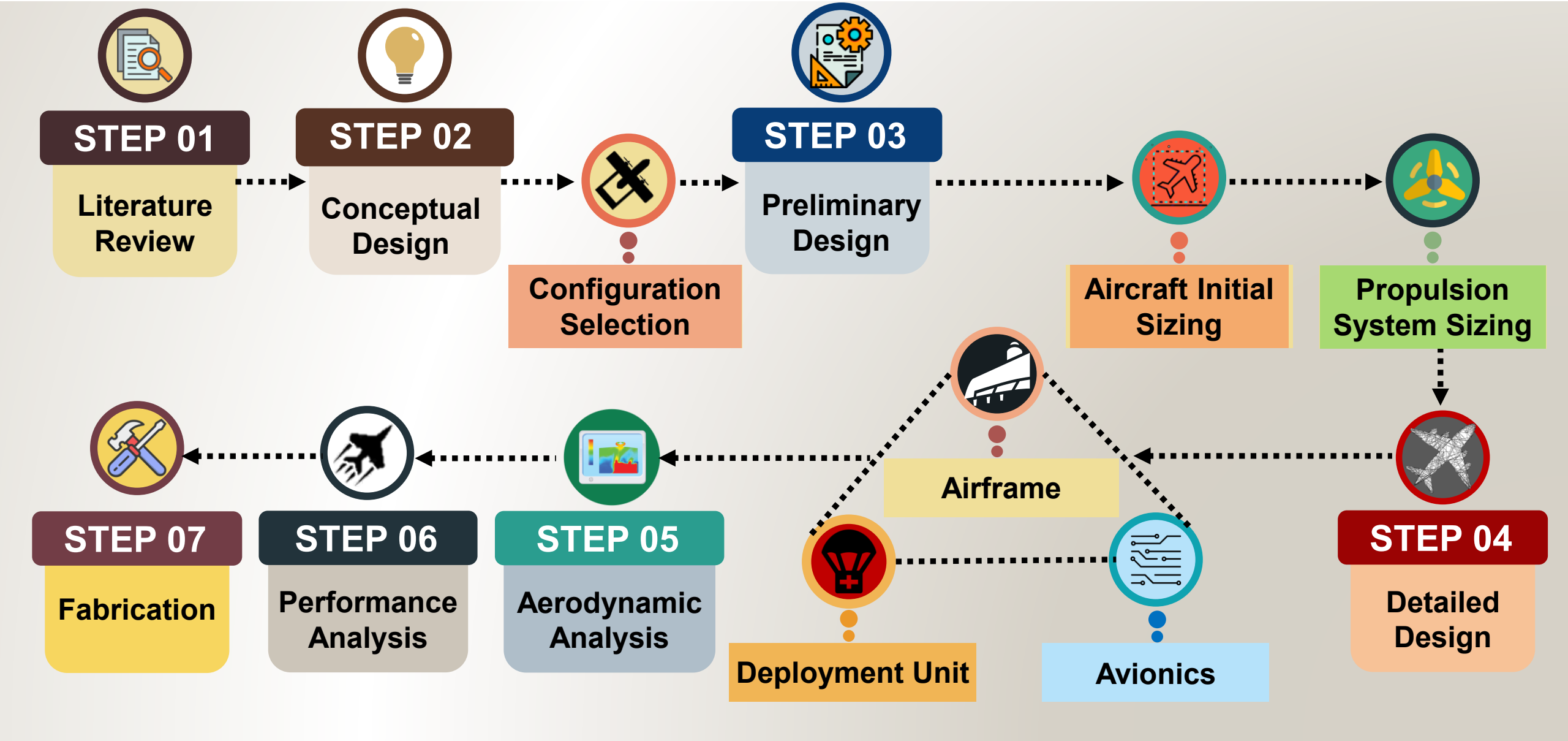
- **Conceptual, Preliminary, and Detailed Design**
- **Estimation of Aerodynamic Efficiency & Performance**
- **Computational Fluid Dynamics Analysis**
- **Design Concept of the Payload Deployment Unit**
- **Fabrication Methodology**



DESIGN METHODOLOGY

SYSTEMS ENGINEERING APPROACH

Project Methodology

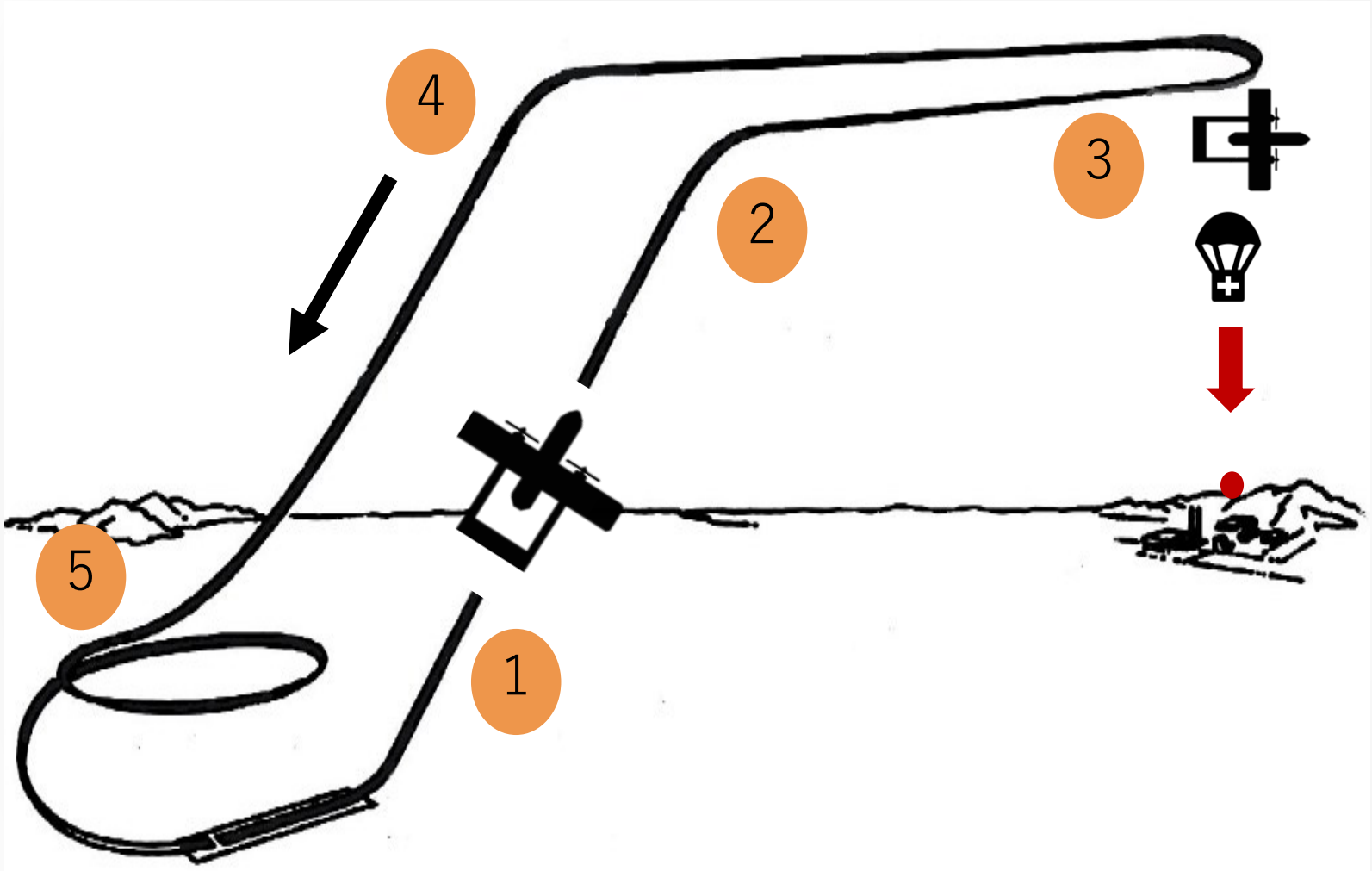


CONCEPTUAL DESIGN

A PLACE TO CONVERGE INNOVATIVE IDEAS



Mission Profile



1. **Take-off from the base**
2. **Cruising to the destination**
3. **Payload dispatched**
4. **Returning to the base**
5. **Loitering before landing**

Design Requirements

FLIGHT REQUIREMENTS	AERODYNAMIC REQUIREMENTS	STRUCTURAL REQUIREMENTS	OTHER REQUIREMENTS
Payload Weight ~ 3 kg	High Lift	Low Empty Weight	Vaccine Box
Range ≤ 800 m	Low Drag	Low Wing Loading	Deployment Ability
Cruise altitude ≥1200 m	Monoplane Wing	Fixed Landing Gear	Radio Controlled
Max. Speed ~ 25 m/s	Tractor Configuration	Balsa & Glass Fiber	Electric Propulsion

Conventional & Twin Boom Configuration: Comparison Study





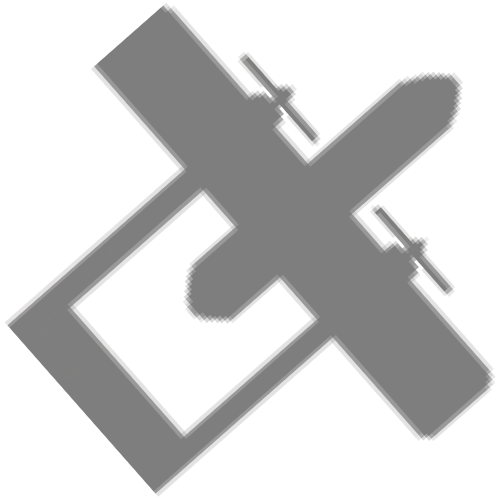
	CONVENTIONAL AIRCRAFT CONFIGURATION	TWIN BOOM MOUNTED AIRCRAFT CONFIGURATION
Payload Bay Volume	The fuselage carries all the avionics systems, making very little room for payloads.	Payloads take up the whole fuselage, with the avionics systems housed in the booms
Empty Weight	Their huge fuselage accounts for increased structural weight.	Structural weight can be reduced significantly.
Wing Loading	Its additional structural weight contributes to low wing loading and affects its flight endurance.	Its inherent low wing loading positively influences its flight endurance.
Stability and Control	Encounters issues with stability as the deployment take place and shifts the C.G.	Exhibits better stability as the rear end of its fuselage remains close to the C.G.

Final Conceptual Design

1. Boom-mounted fixed-wing configuration
2. U-shaped stabilizer system
3. Propeller-based electric propulsion system
4. Double propeller tractor configuration
5. Landing gears in tricycle arrangement



PRELIMINARY DESIGN

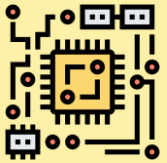
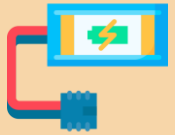
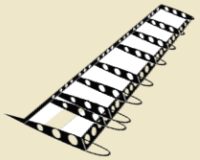


ANALYTICAL COMPUTATION

HISTORICAL DATA

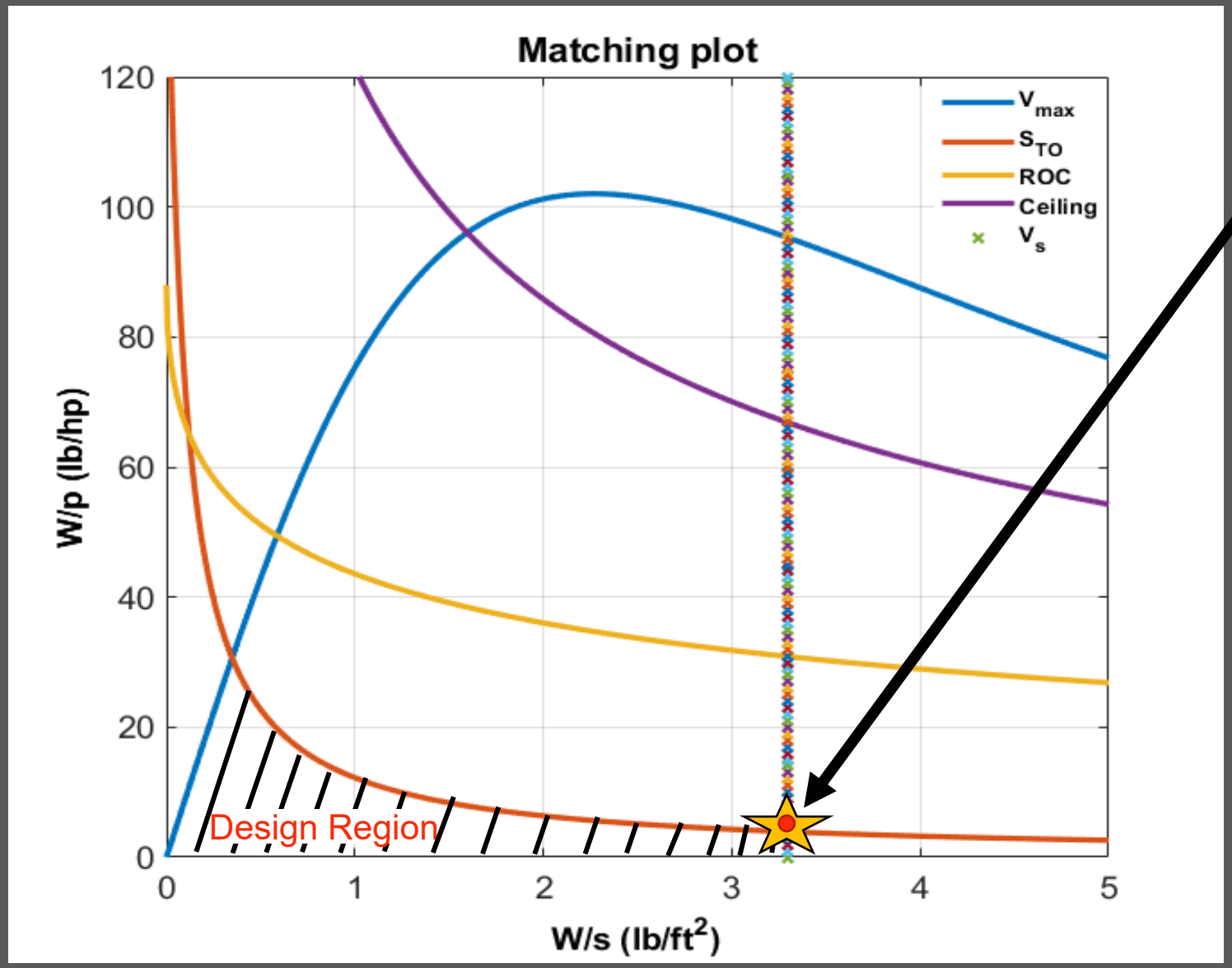
DESIGN SOFTWARE

Aircraft Initial Sizing: Maximum Takeoff Weight

MAJOR WEIGHT CONSTITUENTS	PRIMARY WEIGHT GROUPS		ESTIMATED WEIGHTS
PAYLOADS	Medical Supplies		3 kg
	Onboard Avionics Components		
ONBOARD ENERGY SOURCE	Battery System		0.8 kg
EMPTY WEIGHT	Airframe Structure		6.2 kg
	Propulsion System		

Estimated MTOW = 10kg

Constraint Diagram: Wing Planform Sizing



OPTIMUM DESIGN POINT

The optimum design point shows a minimized **power loading** of **5 lb./hp**, along with a corresponding **wing loading** of **3.3 lb./ft^2**

The obtained wing loading yielded a **wing planform area** of **6.25 ft^2 (0.58 m^2)**

The obtained power loading yielded a **required power level** of **4.41 hp (3.3 kW)**

Propulsion System Sizing



Motor Model: EMAX GT 4030/06 Motor
Number: 02
Available Power: 1.92 kW per motor
Available Thrust: 53 N per motor

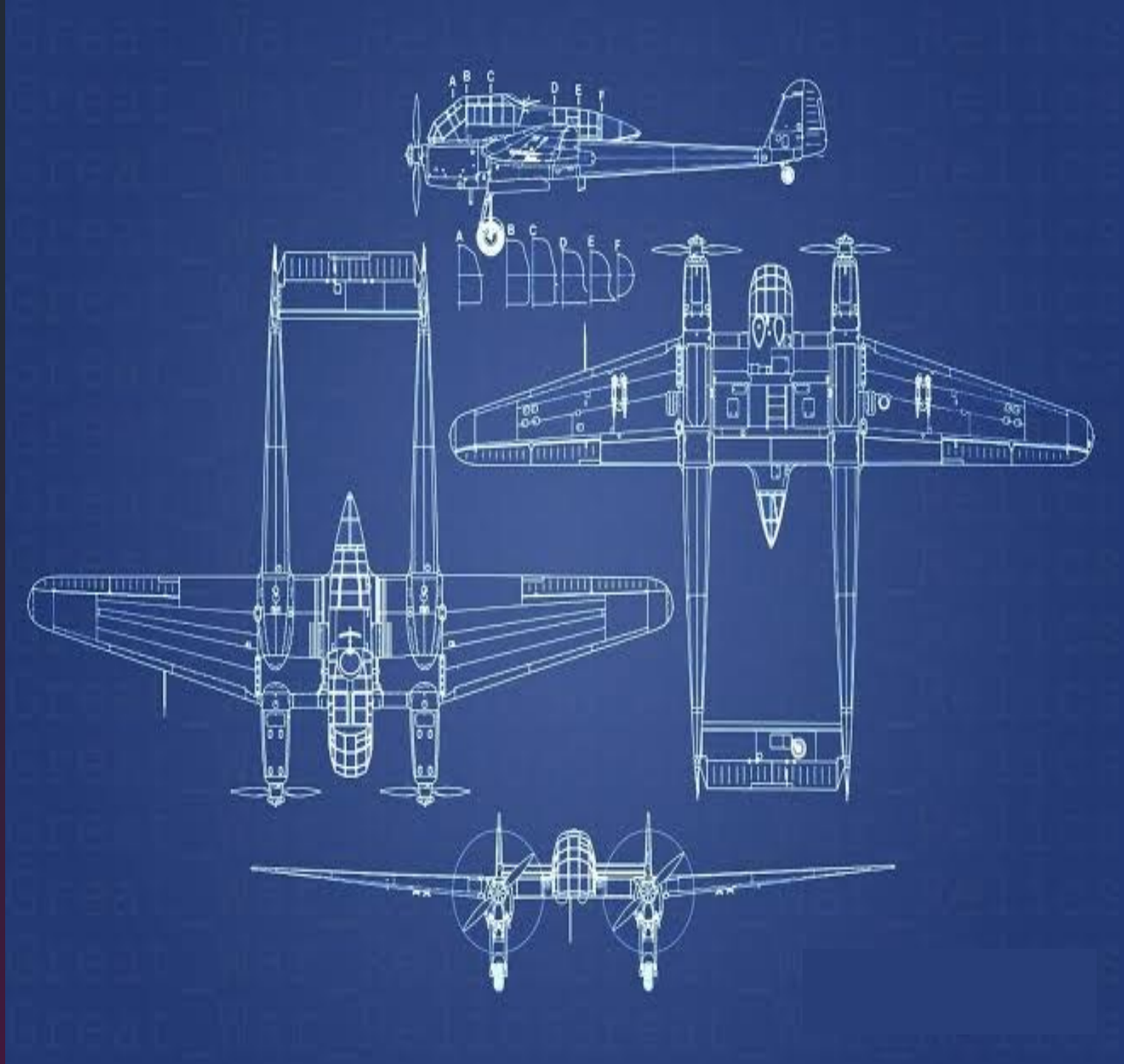
The motors are coupled with a **6S Li-Po Battery** which serves as the onboard energy system.

Propeller Model: Sail 1608 Propeller
Number: 02
Pitch: 8 inches
Propeller Efficiency: 80%

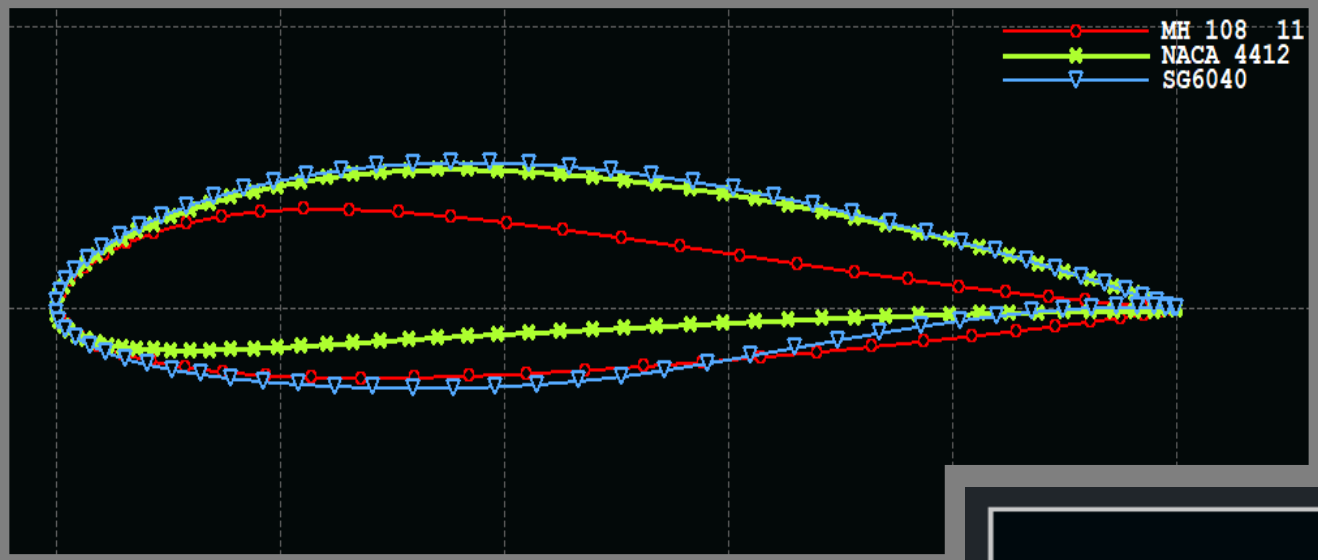
Both CW and CCW types have been used to balance the rotational inertia



DETAILED DESIGN

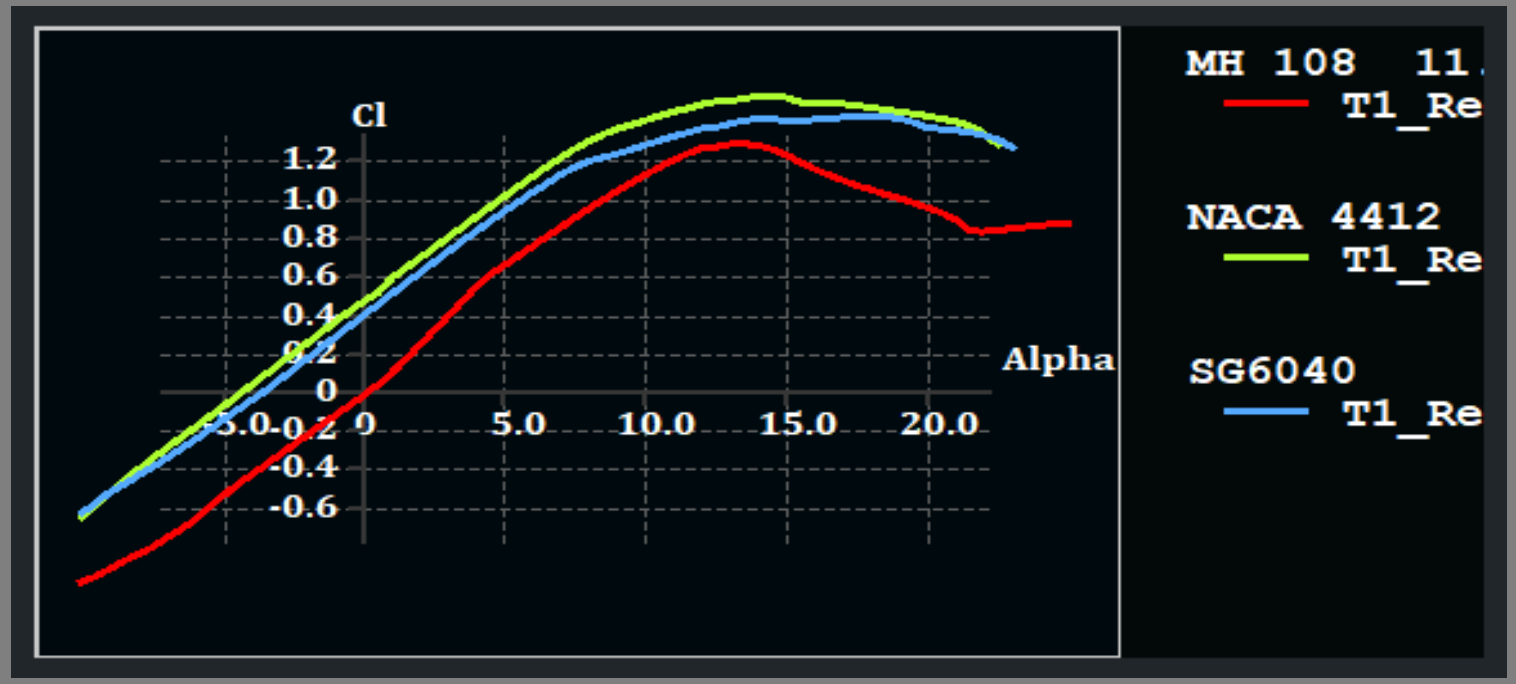


Wing Architecture: Airfoil Selection

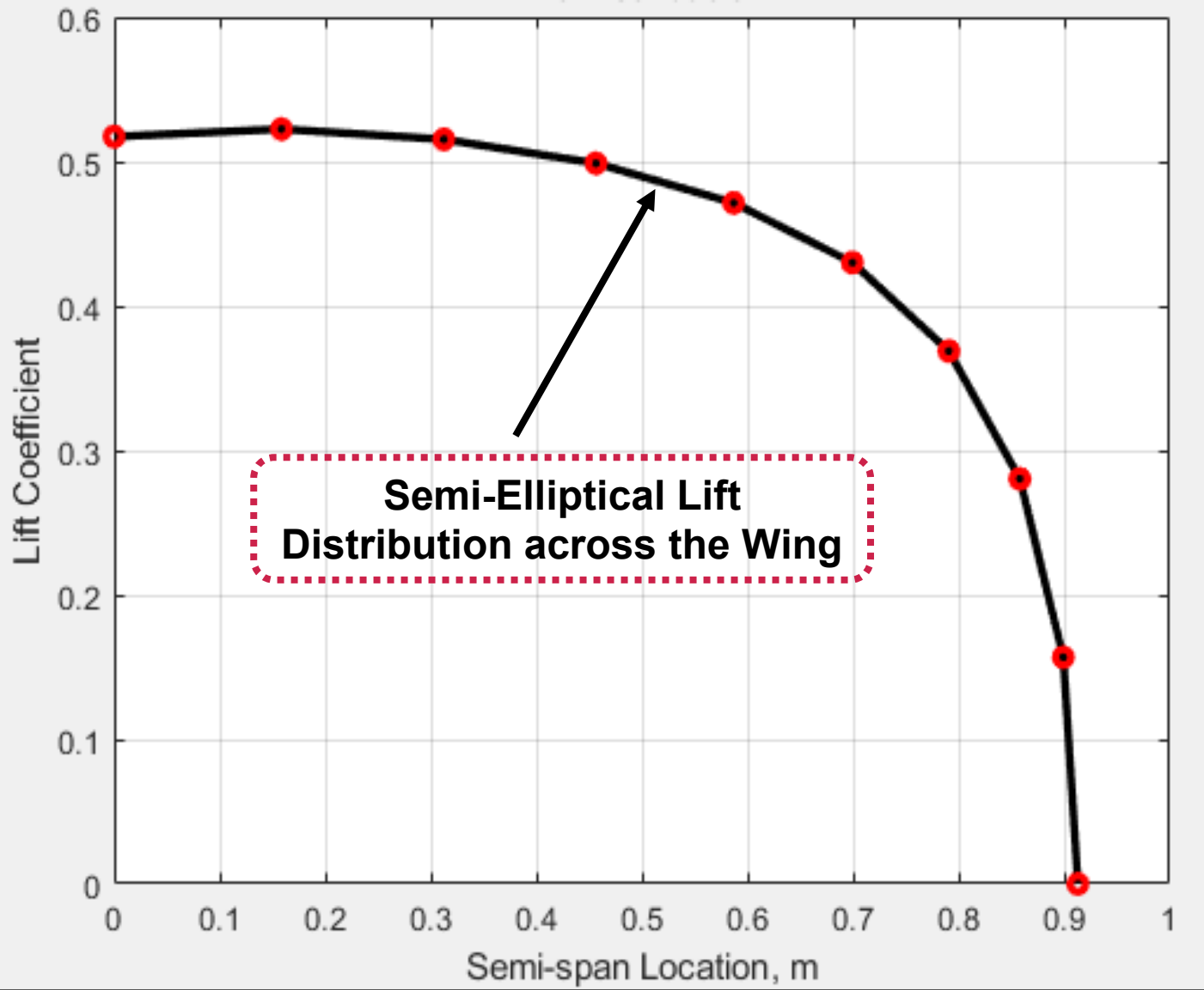


Three airfoil profiles were analyzed prioritizing their aerodynamic properties, namely-
1. NACA 4412
2. MH 108
3. SG 6040

A comparison of the C_L vs. α curves of the three airfoils highlighted the **NACA 4412 airfoil** as the most suitable one.



Wing Architecture: Lift Distribution Curve

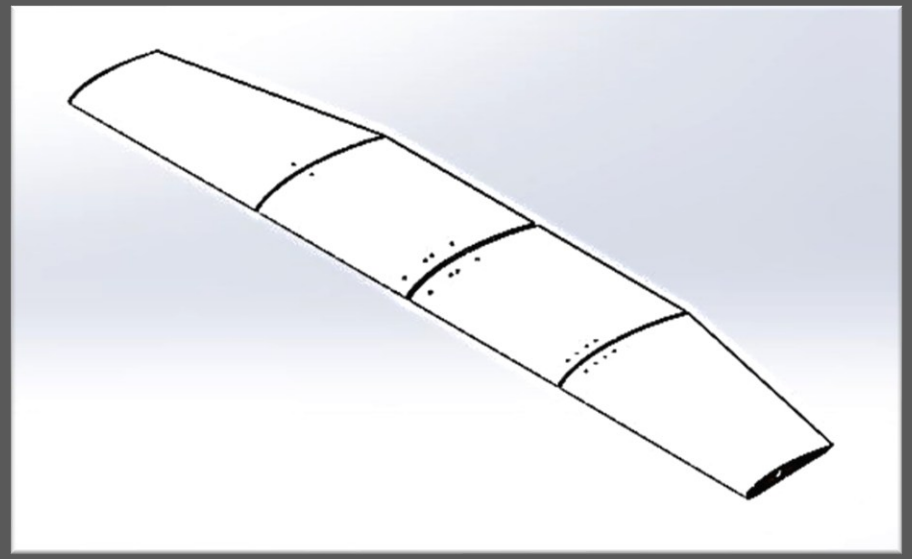


The **Prandtl Lifting Line Theory** was initially introduced to predict the **lift distribution** over the **wing span.**

This analysis was performed iteratively with modifying sets of wing geometry, (such as **spanwise chord distribution, twist, and incident angle**) and **flight conditions** until a semi-elliptical lift distribution was achieved across the wing span.

Finally, a graphical illustration of the **semi-elliptical lift distribution** across the SE-UAV wing was obtained, as shown.

Wing Architecture: Finalized Geometry



Geometric Considerations:

Vertical position relative to the fuselage: High
Wing planform shape: Tapered
Sweep Angle: 0°
Twist angle: -1°
Incidence angle: 2°

Geometric Ratios:

Aspect ratio: 5.75
Tapered ratio: 0.8

Geometric Dimensions:

Wing planform area: 900 in^2 (6.25 ft^2)
Wing span: 72 in (6 ft.)
Root chord length: 13.92 in (1.160 ft.)
Tip chord length: 11.14 in (0.928 ft.)
Mean aerodynamic chord: 12.6 in (1.05 ft.)

Payload Pod Configuration

Special Features:

The entire fuselage was transformed into a streamlined payload capsule to accommodate the deployment unit and the dummy vaccine boxes.

Geometric Dimensions:

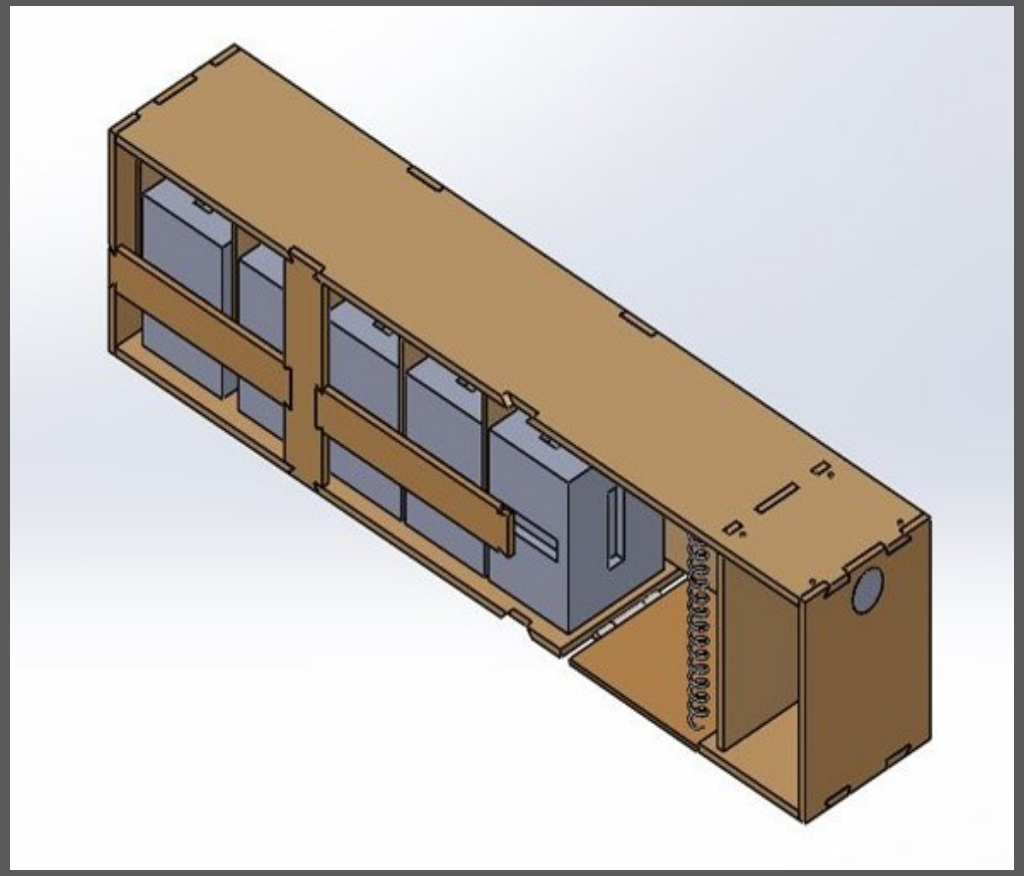
Capsule length: 44.4 in (3.7 ft.)

Width: 3.96 in (0.33 ft.)

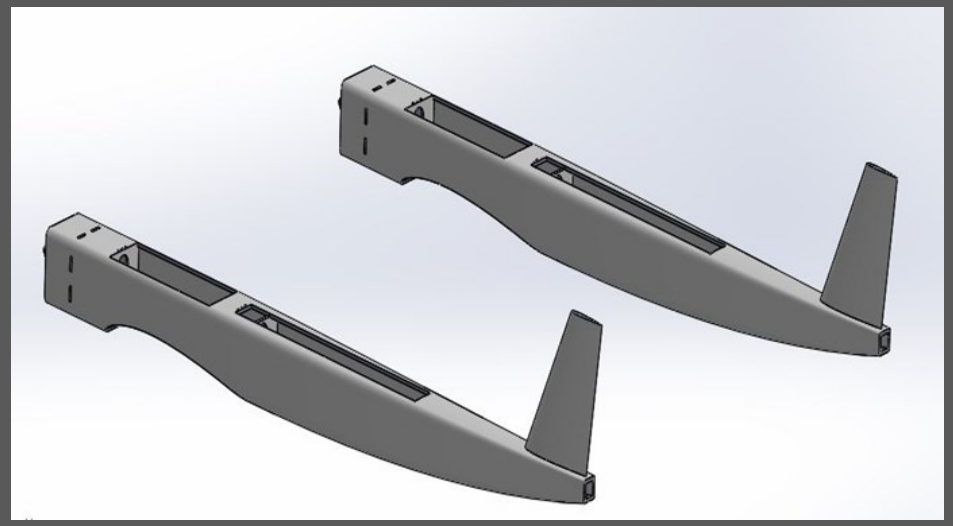
Height: 5.12 in (0.427 ft.)

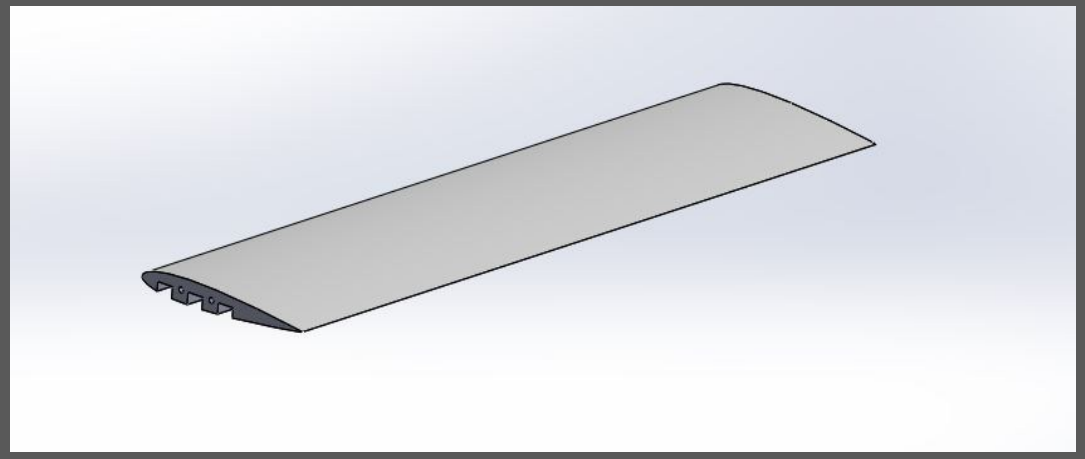
Volume: 864 in^2 (0.5 ft^2)

Slenderness ratio: 8.8



Boom-mounted Empennage Lay-out





Geometric Considerations:

Empennage Configuration: U-shaped boom-mounted empennage system

Airfoil: NACA 0012

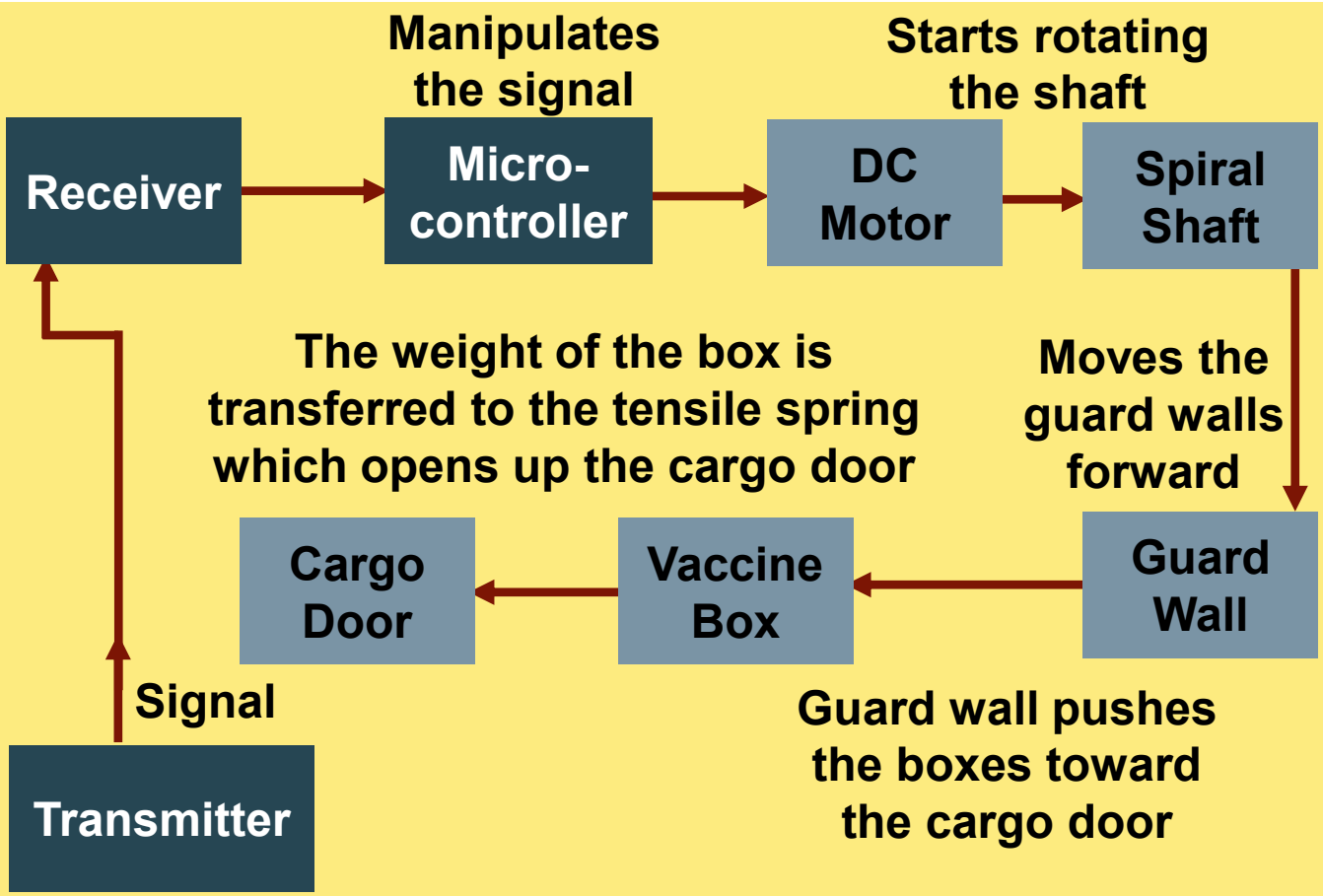
Design Features of Vertical Stabilizers:

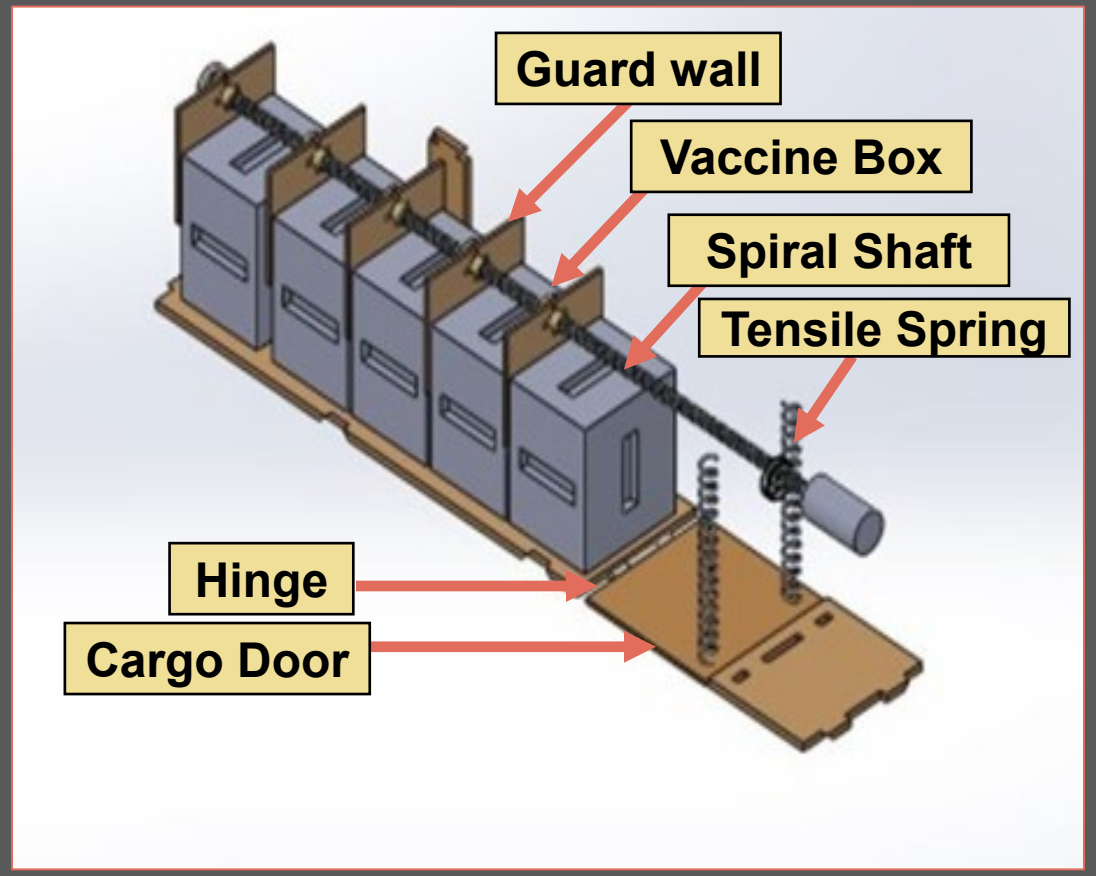
Aspect ratio: 1.75
Wing span: 8.04 in (0.67 ft.)
Planform area: 37 in^2(0.256 ft^2)
Taper ratio: 0.5

Design Features of Horizontal Stabilizer:

Aspect ratio: 4.28
Wing span: 32 in (2.65 ft.)
Planform area: 236.6 in^2 (1.643 ft^2)
Taper ratio: 1
Chord length: 7.32 in (0.61 ft.)

Deployment Mechanism

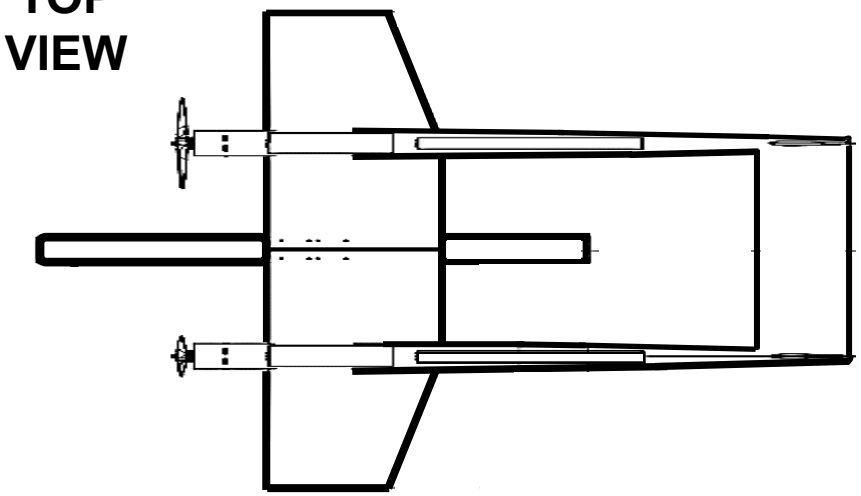




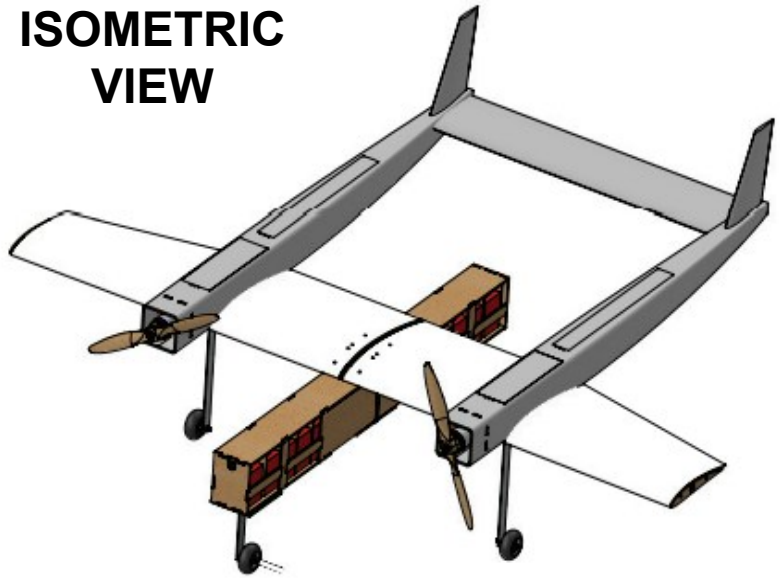
After dropping each vaccine box, the motor is stopped by cutting off the signal which brings the springs back to their initial position and closes the door.

Geometry Modeling

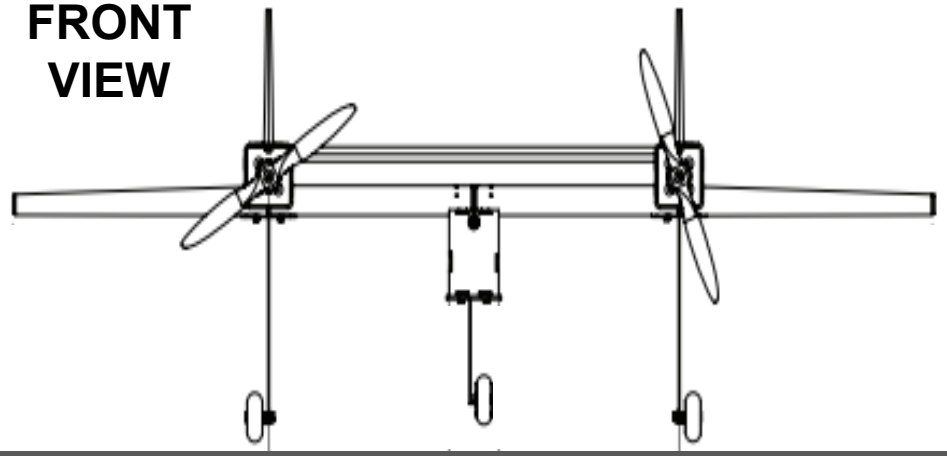
TOP VIEW



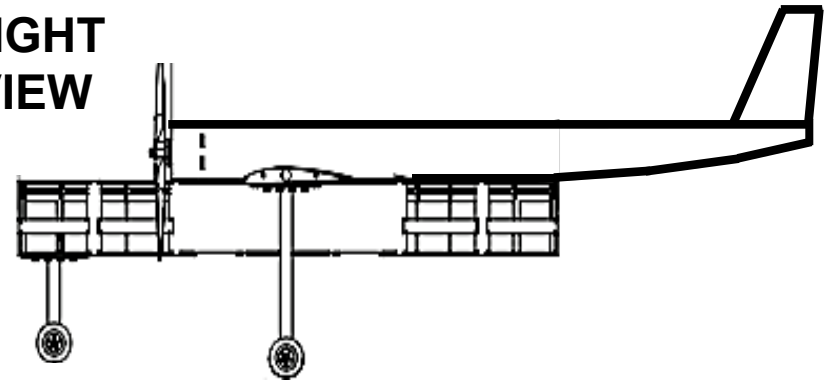
ISOMETRIC VIEW



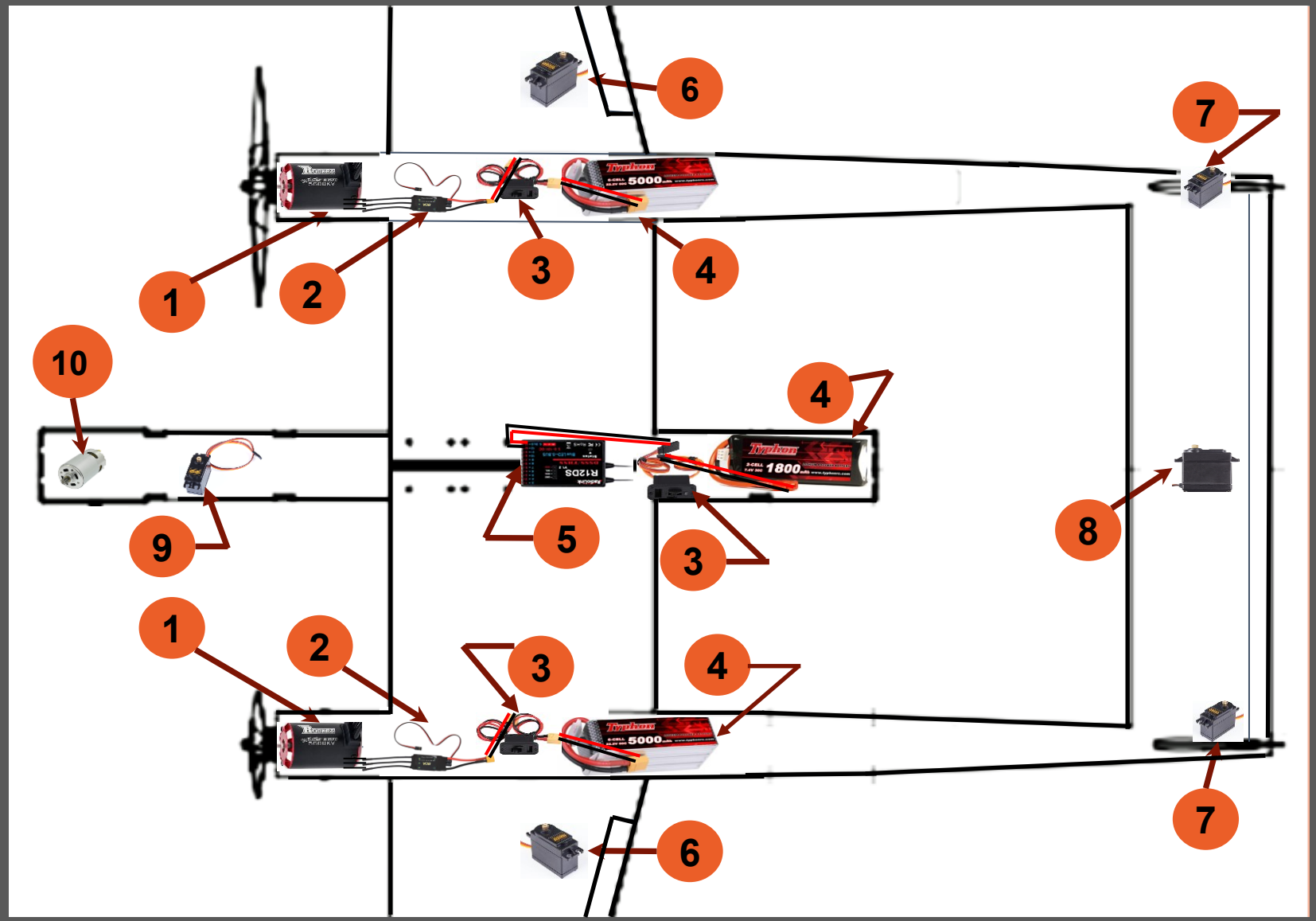
FRONT VIEW



RIGHT VIEW

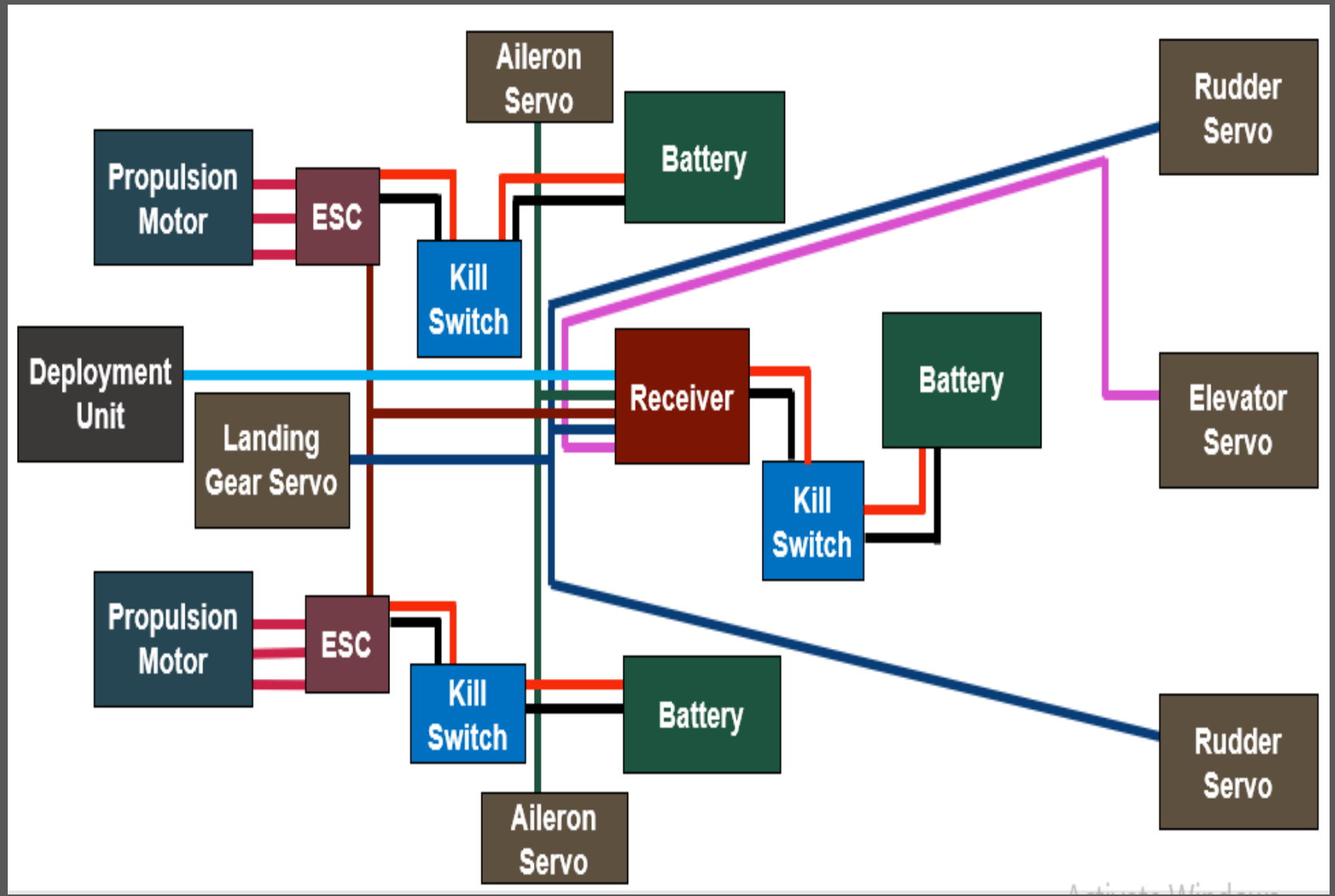


Avionics System Layout



1. Motor
2. ESC
3. Kill Switch
4. Battery
5. Receiver
6. Aileron Servos
7. Rudder Servos
8. Elevator Servo
9. Nose Gear Servo
10. Deployment Unit

Avionics System Circuit Design



Propulsion Motor Connection

Throttle Signal

Battery +ve end

Battery -ve end

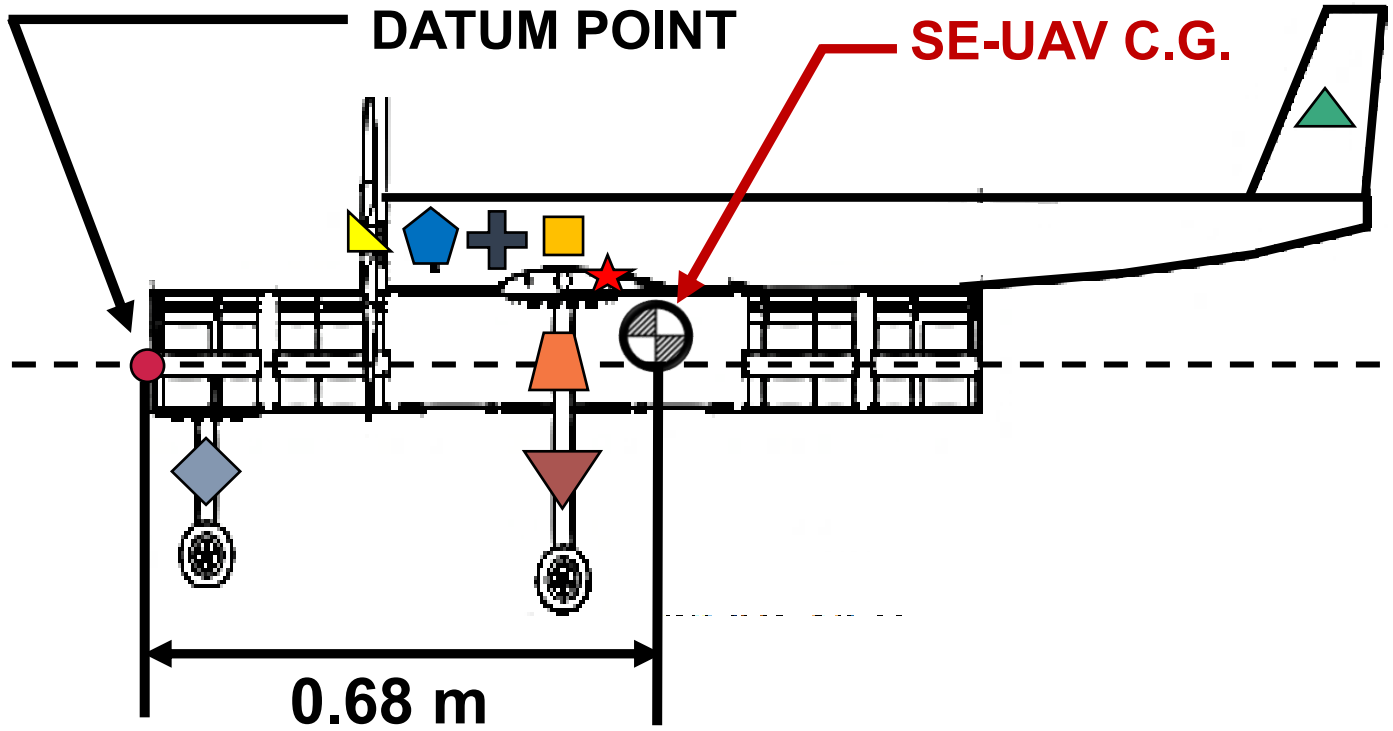
Aileron Servo

Rudder & Landing Gear Servo

Elevator Servo

Deployment Unit

Final Weight Distribution and the Centre of Gravity



- Wing
- Payload Pod with Deployment Unit
- Boom-mounted Vertical Tail
- ESC
- Motor
- Propeller
- Battery
- Nose Gear
- Main Gear

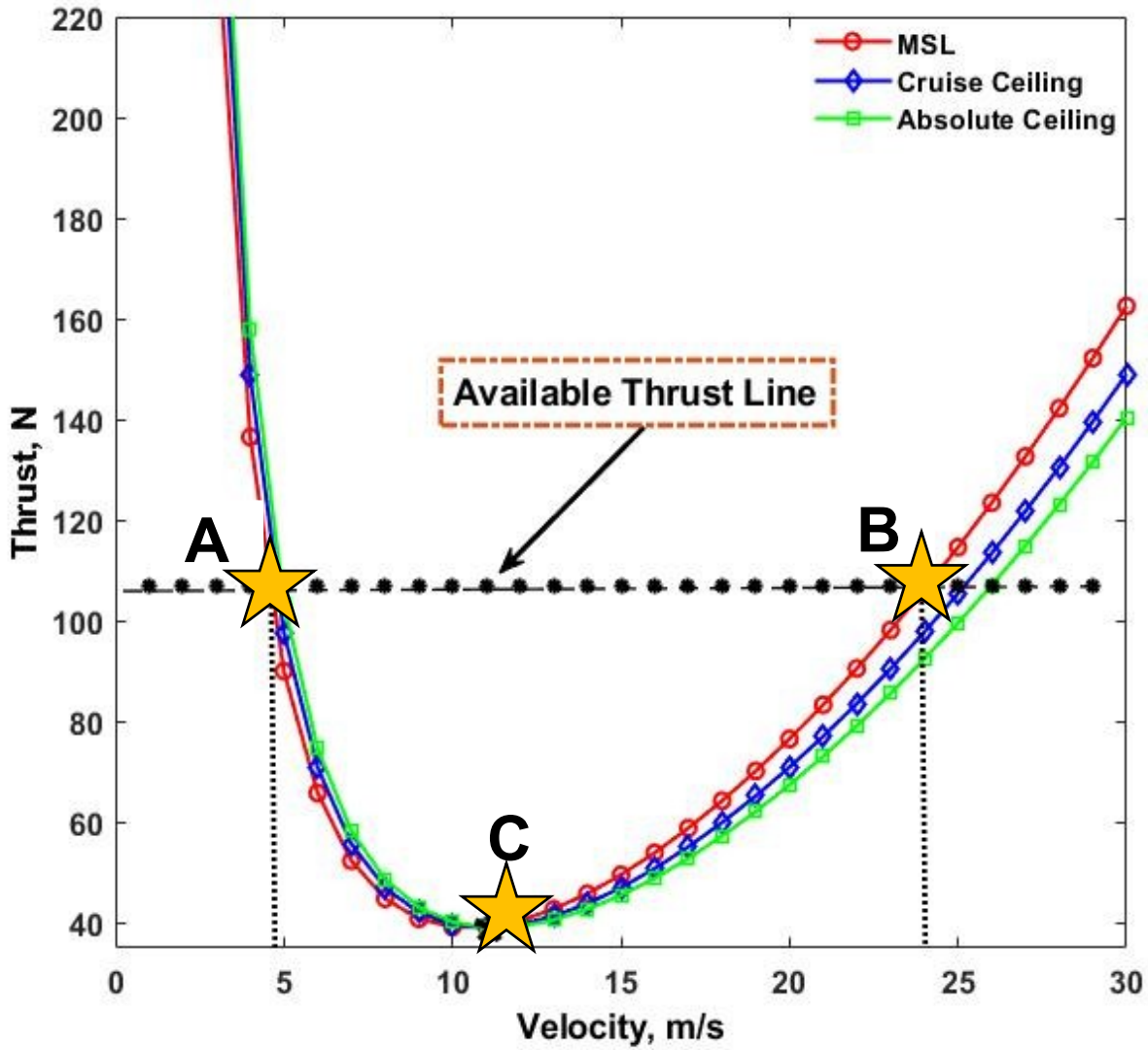
The position of the SE-UAV Centre of Gravity has been determined using,

$$(X, Y)_{C.G.} = \frac{\sum w.\, d_{(x,y)}}{\sum w}$$

The individual weights (w) of each aircraft component were determined from the **"Mass Properties"** feature of **SOLIDWORKS**, along with their distances (d) from the datum point.

The C.G. was found at **0.68 m distance from the front wall of the payload capsule**, which is approximately **17% of the mean aerodynamic chord**.

Performance Analysis: Drag Polar Curve

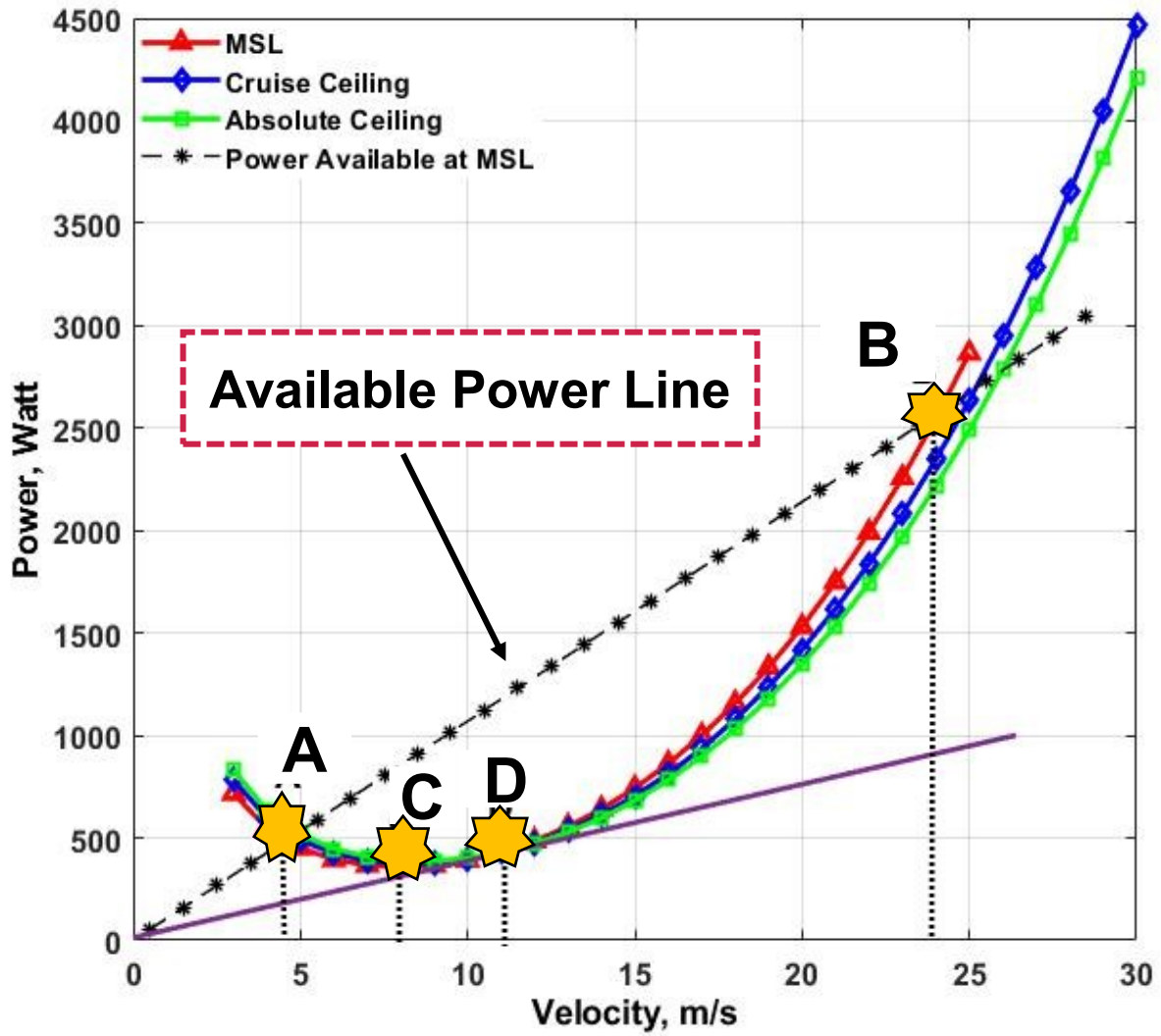


Point A highlights the **stall speed** of the aircraft which is approximately **5 m/s**.

Point B shows the **maximum speed** that the SE-UAV can attain at MSL without compromising its steady-level flight conditions, which is **24 m/s**.

Point C represents the **maximum velocity** at which the aircraft will require the **least amount of thrust** at MSL, which is **11 m/s**.

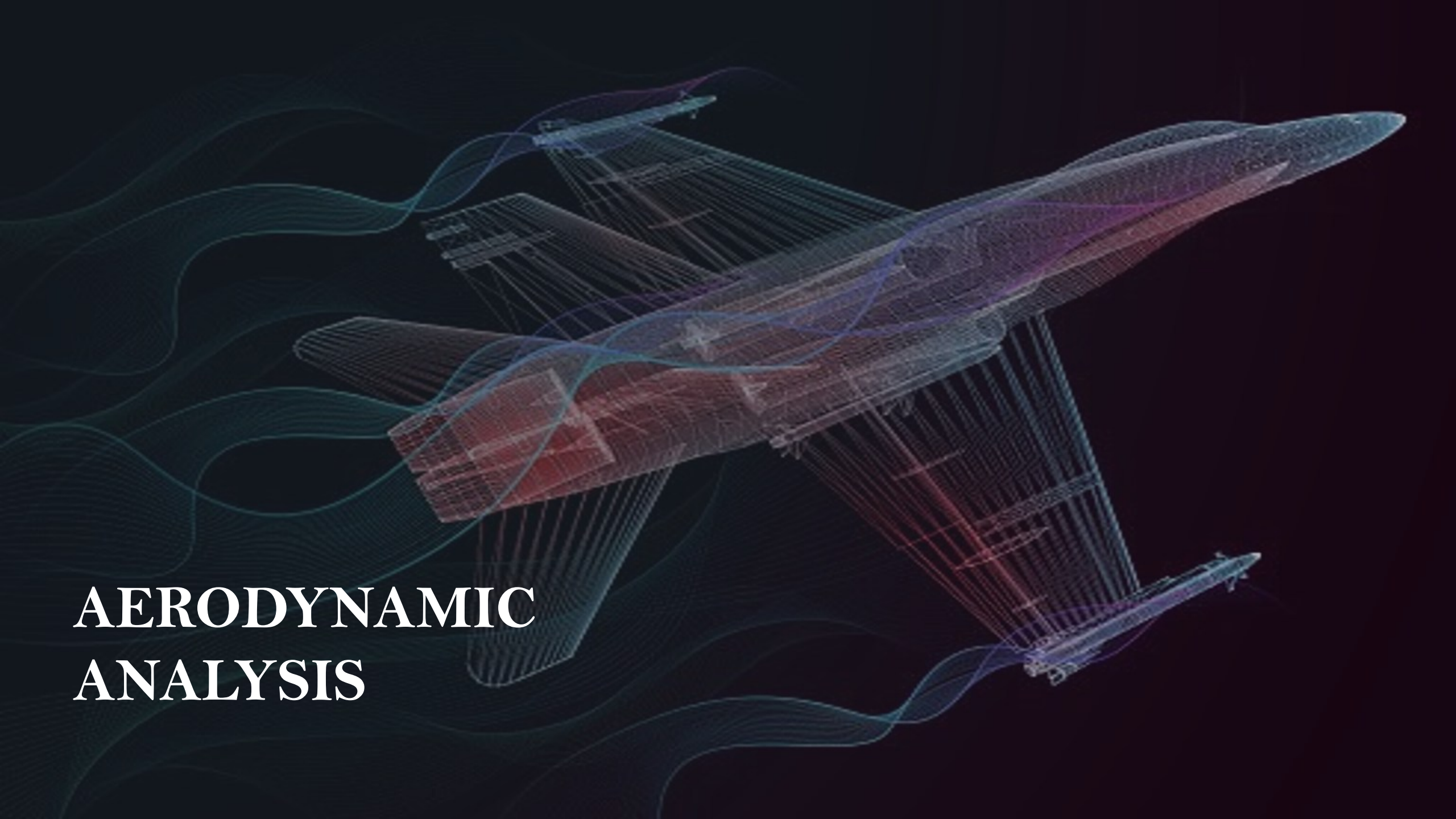
Performance Analysis: Required Level of Power



The available power line at the MSL also constrained the **operational speed** limit from **5-24 m/s**, as highlighted by **points A** and **B.**

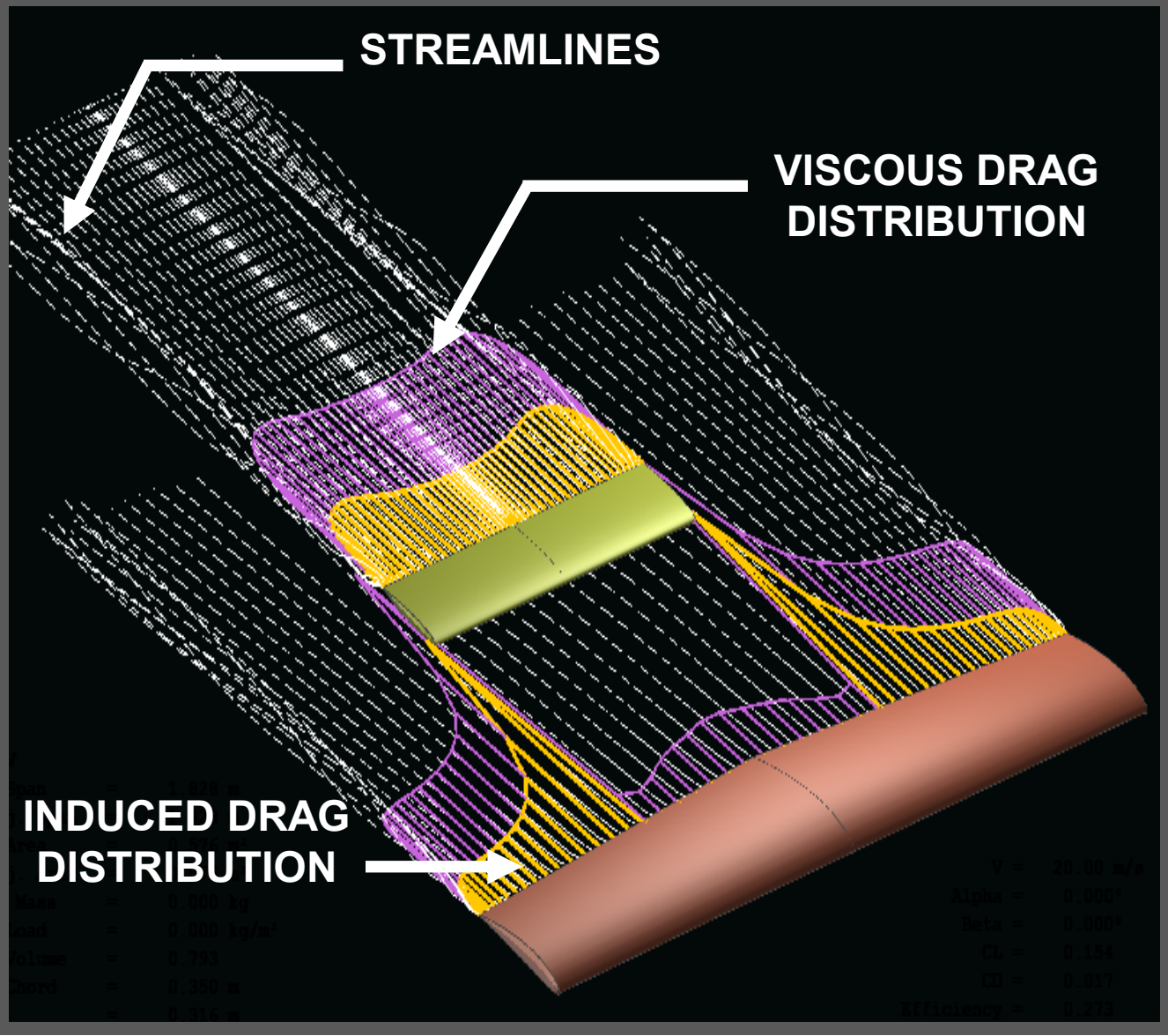
Point C shows the **maximum endurance speed** of the SE-UAV at MSL, which was found to be **8 m/s**.

Point D represents the **maximum range speed** which is equal to **11 m/s**.



AERODYNAMIC ANALYSIS

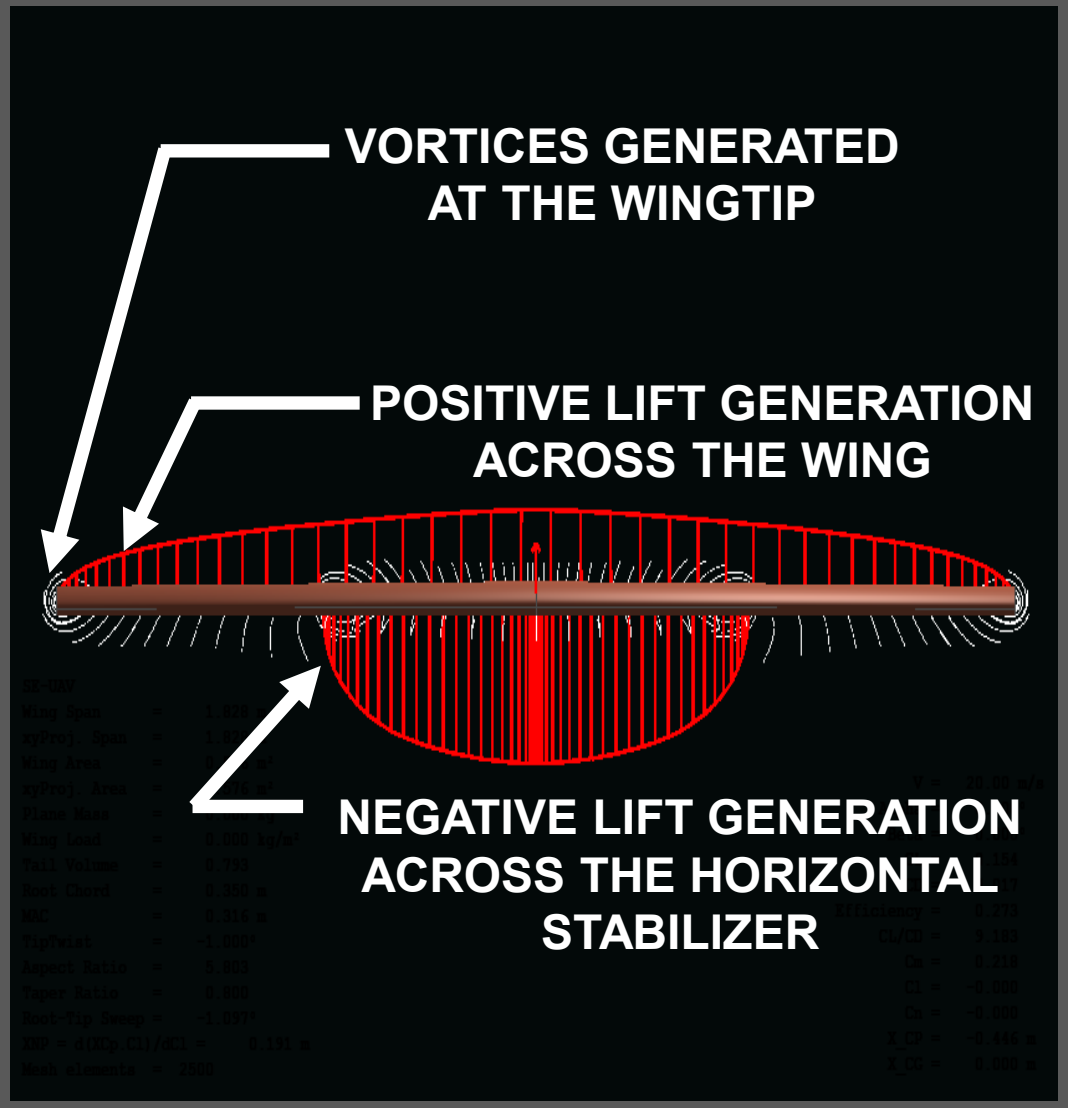
Induced Drag Distribution over the Lifting Surfaces

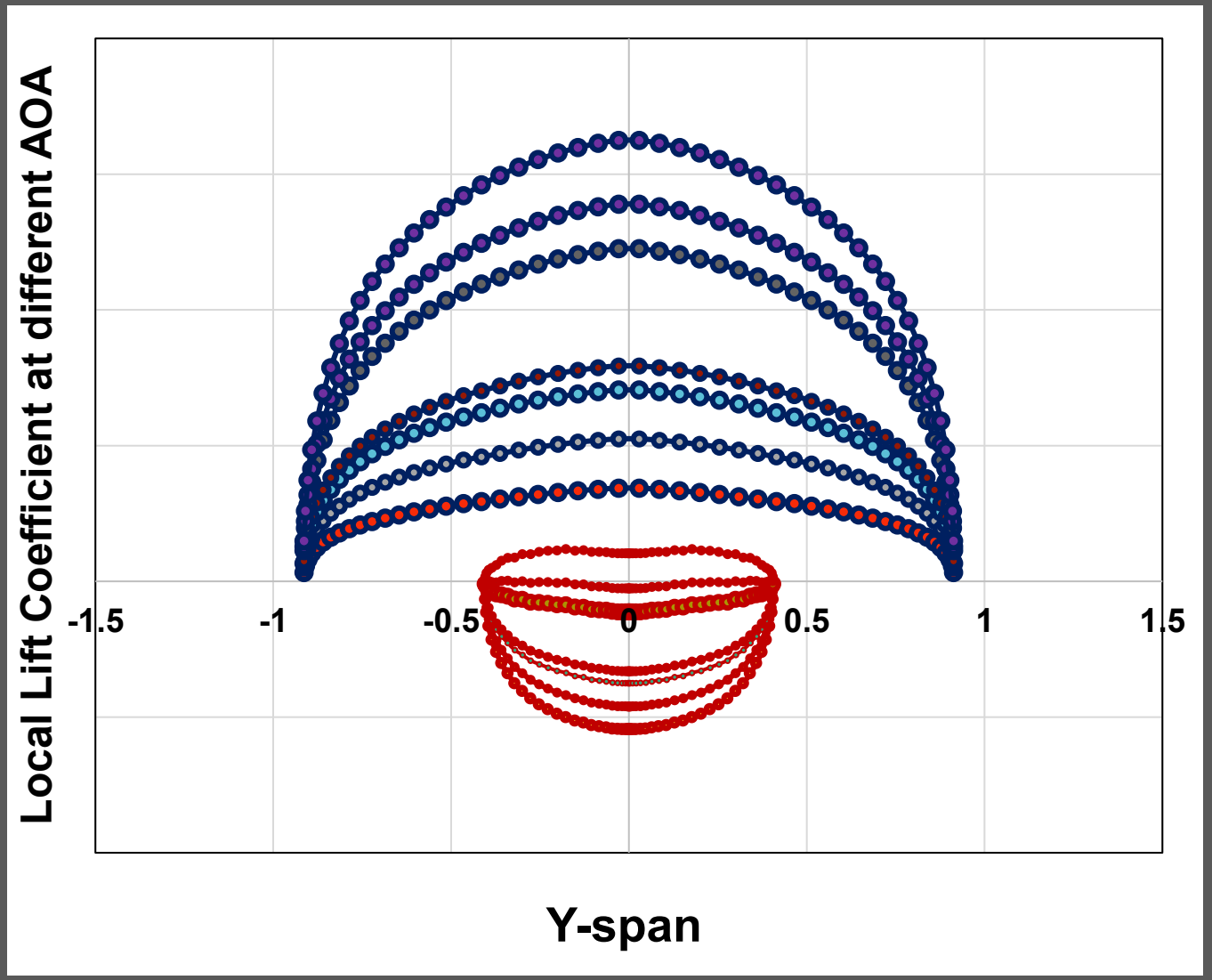


The **Vortex Lattice Method** was introduced as a low-fidelity methodology for analyzing the **induced drag and lift distribution** of the **wing** and the **horizontal stabilizer**.

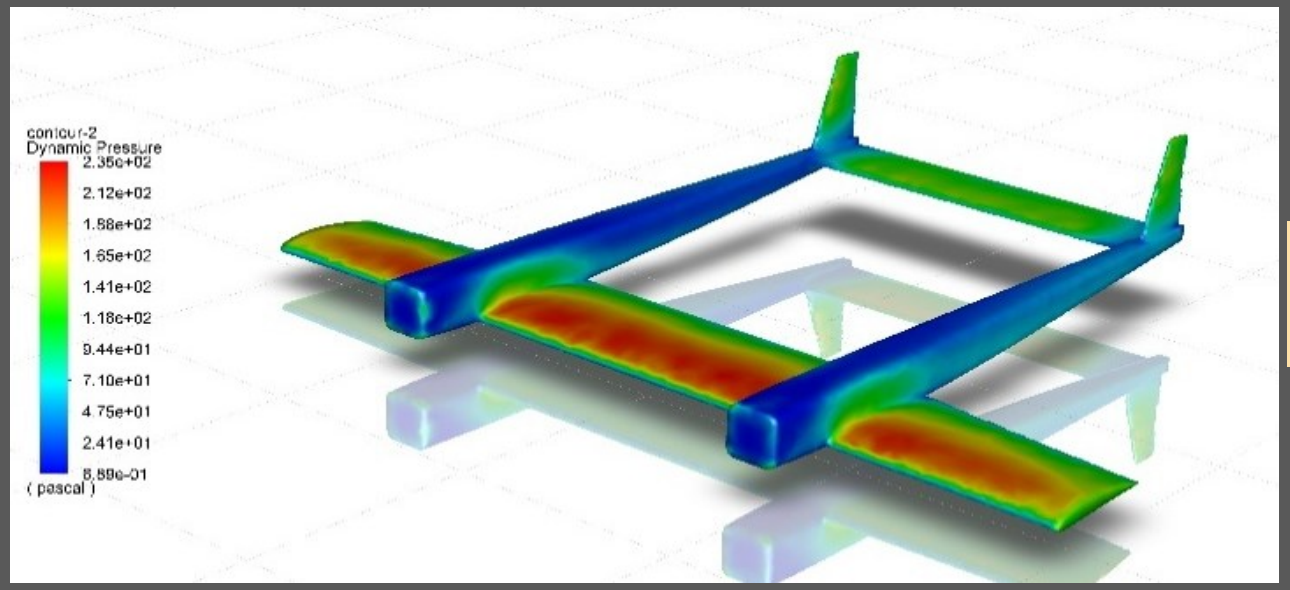
This analysis was performed by generating **a 3-D model of the lifting surfaces**, followed by the analysis of their **2-D airfoils** for a wide range of **Reynold's numbers** and **angles of attacks.**

Lift Generation across the Lifting Surfaces



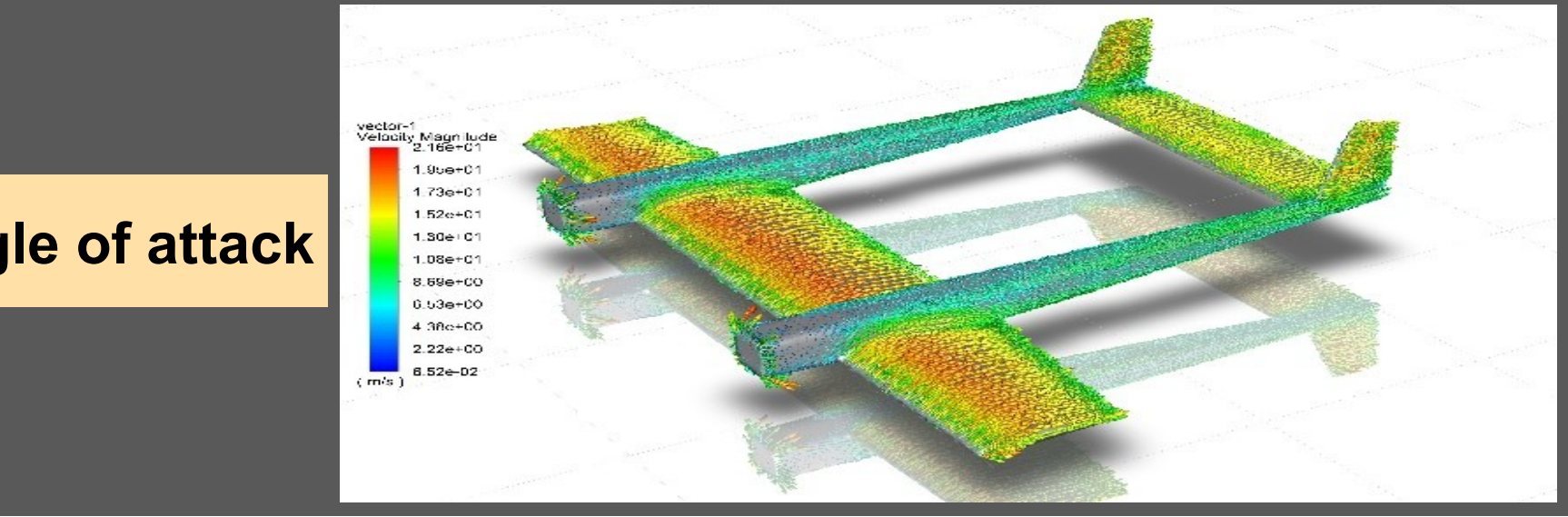


CFD Analysis of the SE-UAV Model

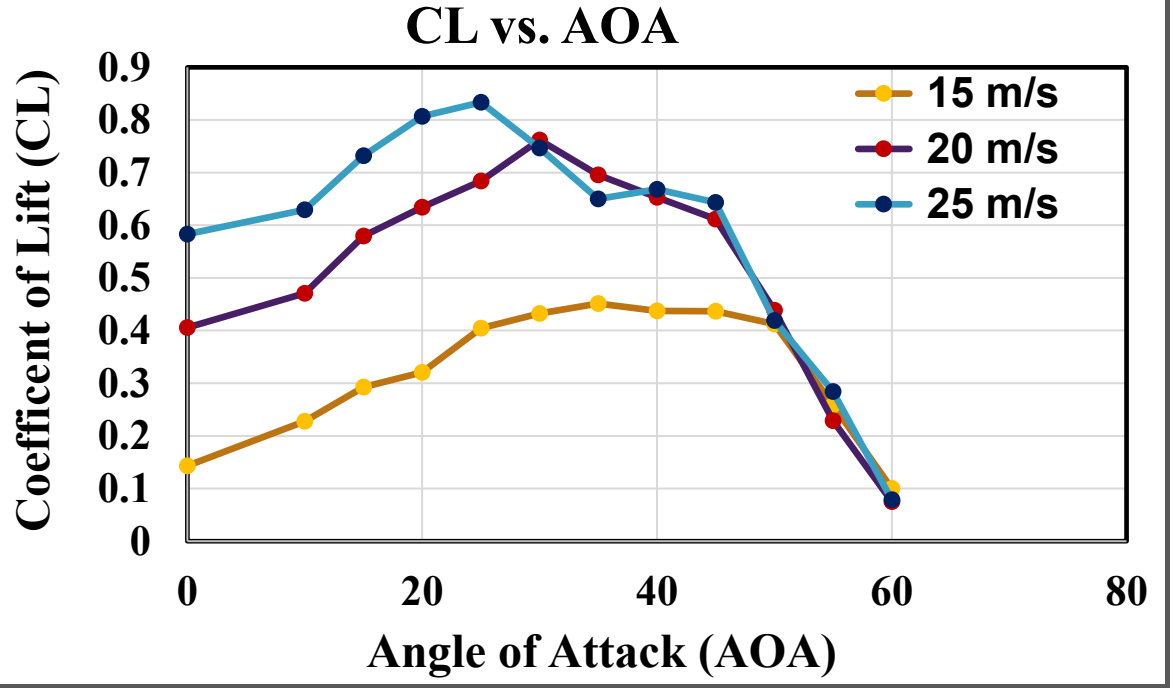


Pressure contour at 0° angle of attack

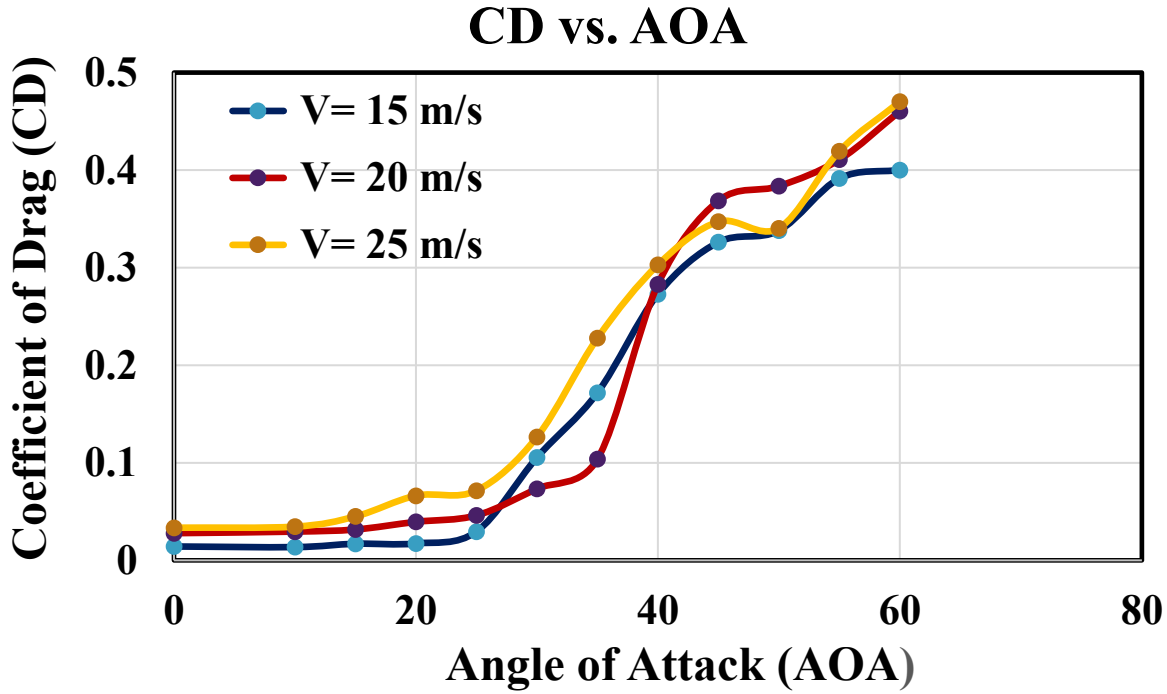
Velocity vector at 0° angle of attack



Outcomes of CFD Analysis



The C_L **vs. AOA** curves for three different airspeeds show that the UAV will be capable to attain a maximum lift coefficient while maintaining a maximum flight speed of 25 m/s at cruise.



Meanwhile, the C_D **vs. AOA** curves show a significant rise in the total drag as the aircraft flies at 25 m/s. Therefore, maintaining **a cruise speed of 20 m/s** was found to be most **suitable**.

FABRICATION



Materials Selection & Manufacturing Process

MAJOR COMPONENTS	MATERIALS USED	MANUFACTURING PROCESSES
WING	• Balsa Wood • Foam • Aluminium Spurs • Glass Fiber • Epoxy Resin • Hardener	• Laser Cutting • Hot-Wire Cutting
EMPENNAGE		
PAYLOAD POD	• Balsa Wood • Epoxy Hardener	• Laser Cutting
BOOM	• Balsa Wood • Plywood	• Laser Cutting
LANDING GEAR	• Stainless Steel Bar • Plywood Plates	• Laser Cutting • Wielding



SE-UAV Fabrication









Final Specifications of SE-UAV

Parameters	
Maximum Take-off Weight	10 kg
Payload Weight	3 kg
Wing Span	1.83 m^2
Maximum Power Available at MSL	3.84 kW
Maximum Speed	24 m/s
Cruise Speed	20 m/s
Stall Speed	5 m/s
Endurance Speed	8 m/s
Endurance	8 mins



Conclusion

- SOARING EAGLE UAV has been developed as a test-model to dispatch dummy vaccine components at a short range. However, this work can be used as a baseline concept for the development of UAV prototypes at a larger scale in the future.
- Besides, once the SE-UAV has its test flight, a new study on the design optimization can be conducted with a view to bringing improvements to the existing design.